

Laboratory 6 — Reliability (GameSeeker)

Course: Software Architecture · 2026-I **Project:** GameSeeker — distributed game-deal discovery platform

This document is the technical guide for Lab 6. It covers **Part A — Cluster Pattern** (Kubernetes) and **Part B — Active Redundancy / Hot Spare**, both implemented as real contributions to the GameSeeker project.

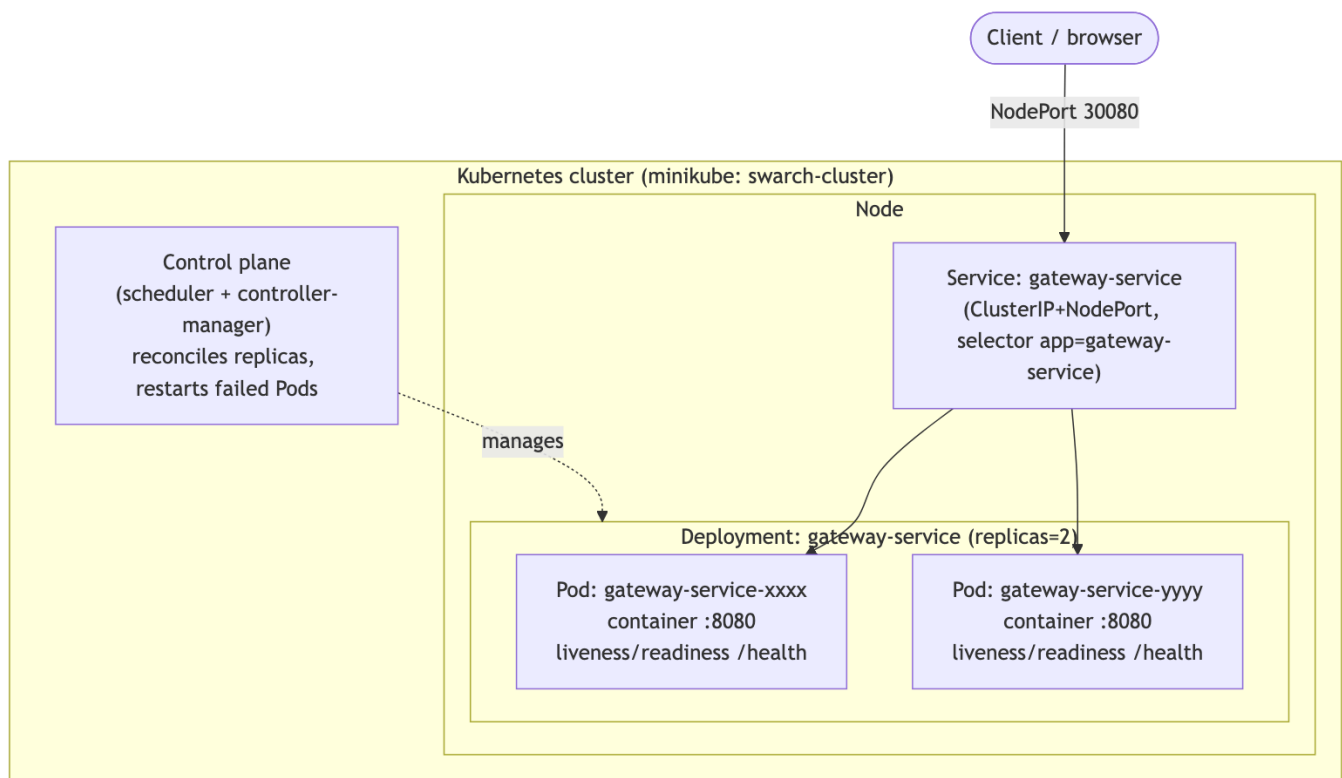
1. Team Information

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2. Architectural Views

2.1 Cluster Pattern (Part A) — Deployment view

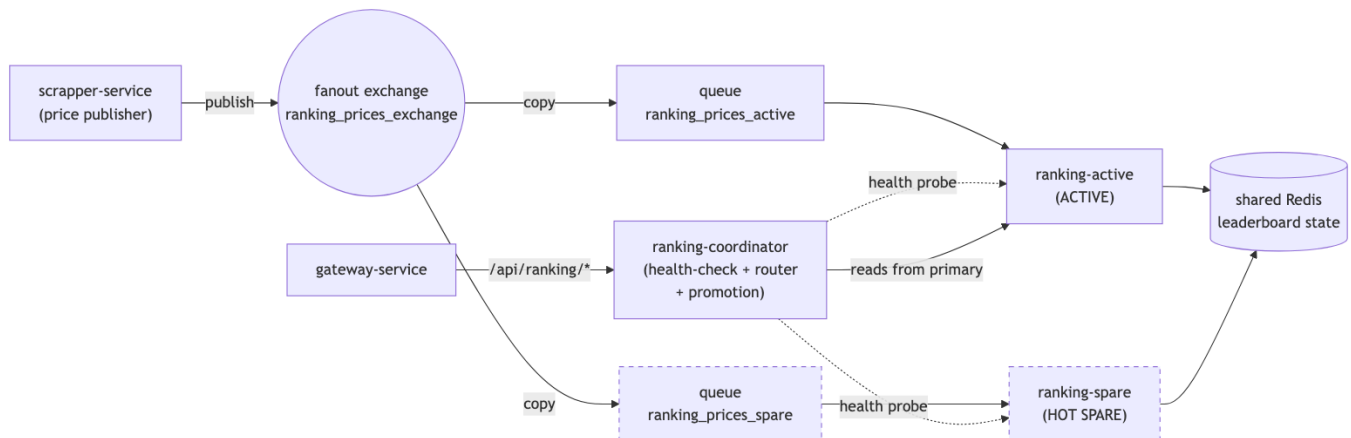
The **gateway-service** (the system's single entry point / orchestrator) is deployed as a Kubernetes Deployment of **2 replicas** behind one Service. Clients reach a single stable endpoint; kube-proxy load-balances across the ready Pods. The control plane continuously reconciles desired vs. actual state, restarting or rescheduling Pods on failure.



New infrastructure introduced: a local Kubernetes cluster (minikube), the `gateway-service` Deployment, and the `gateway-service` Service (`k8s/gateway-deployment.yaml` , `k8s/gateway-service.yaml`).

2.2 Active Redundancy / Hot Spare (Part B) — C&C view

The **ranking-service** is duplicated into an **active** and a **hot spare** node. A RabbitMQ **fanout exchange** delivers every price event to *both* nodes, so both keep fully synchronized leaderboards (state in shared Redis). A **coordinator** health-checks both nodes, serves leaderboard reads from the active, and promotes the spare the instant the active fails.



New infrastructure introduced: the fanout exchange `ranking_prices_exchange` , a second ranking node (`ranking-spare`), and the `ranking-coordinator` (`reliability/ranking-coordinator/`). State synchronization mechanism = **parallel consumption of the fanout exchange + shared Redis**.

3. Technical Guide — Part A: Cluster Pattern

3.1 Pattern description

The **Cluster Pattern** groups multiple identical nodes under a unified management layer so they behave as a single logical machine. Kubernetes implements it natively: containers run in **Pods**, Pods are grouped by a **Deployment**, and a **Service** exposes them behind one stable endpoint. The control plane supplies the reliability tactics:

- **Fault detection** — liveness/readiness probes on `/health` .
- **Recovery / redundant spare** — the Deployment keeps `replicas` Pods alive; a deleted or crashed Pod is recreated automatically.
- **Load balancing** — the Service spreads traffic across ready Pods.

3.2 Implemented cluster type

Active/Active. Every gateway replica serves traffic simultaneously. We chose Active/Active because the gateway is **stateless** (it only routes/validates and holds no session state), so there is no benefit to keeping a node idle: running all replicas hot maximizes both throughput and fault tolerance, and losing a Pod never loses state. This contrasts with the Active/Passive model we use for the ranking node in Part B, where a designated spare stands by behind a coordinator.

3.3 Implementation steps

1. Add a hostname-reporting `/health` endpoint to the gateway so probes have a target and load-balancing across replicas is observable (`gateway-service/src/index.ts`).
2. Author the Kubernetes manifests (`k8s/gateway-deployment.yaml` , `k8s/gateway-service.yaml`).
3. Start a local cluster: `minikube start -p swarch-cluster` .
4. Build the gateway image straight into minikube's Docker daemon (`eval $(minikube -p swarch-cluster docker-env) → docker build`).
5. `kubectl apply -f k8s/gateway-deployment.yaml -f k8s/gateway-service.yaml` .
6. Demonstrate self-healing (`kubectl delete pod ...`) and scaling (`kubectl scale deployment/gateway-service --replicas=N`).

All of the above is automated in `k8s/demo.sh` .

3.4 Configuration snippets

Deployment (project-specific adaptations highlighted in comments — 2 replicas, probes on the gateway's real `/health` , and the three env vars `gateway-service/src/lib/env.ts` requires at boot):

```
# k8s/gateway-deployment.yaml (excerpt)
spec:
  replicas: 2
  template:
    spec:
      containers:
        - name: gateway-service
          image: gamesseeker/gateway-service:lab6
          ports: [{ containerPort: 8080 }]
          readinessProbe: { httpGet: { path: /health, port: 8080 } }
          livenessProbe: { httpGet: { path: /health, port: 8080 } }
```

```
# k8s/gateway-service.yaml (excerpt)
spec:
  type: NodePort
  selector: { app: gateway-service }
  ports: [{ port: 80, targetPort: 8080, nodePort: 30080 }]
```

3.5 Evidence — self-healing and scaling

Captured from `k8s/demo.sh` against minikube (`swarch-cluster`).

Deployment up — 2 replicas behind one Service:

```
$ kubectl get pods -l app=gateway-service -o wide
```

NAME	READY	STATUS	RESTARTS	AGE	IP	NODE
gateway-service-85bd46cc4c-46qf8	1/1	Running	0	7s	10.244.0.3	swarch-cluster
gateway-service-85bd46cc4c-7zkh5	1/1	Running	0	7s	10.244.0.4	swarch-cluster


```
$ kubectl get svc gateway-service
```

NAME	TYPE	CLUSTER-IP	EXTERNAL-IP	PORT(S)	AGE
gateway-service	NodePort	10.104.253.64	<none>	80:30080/TCP	7s

Self-healing — delete a Pod, the Deployment recreates it automatically:

```
$ kubectl delete pod gateway-service-85bd46cc4c-46qf8
pod "gateway-service-85bd46cc4c-46qf8" deleted
```



```
$ kubectl get pods -l app=gateway-service      # count restored to 2 (new Pod -fh7nf)
```

NAME	READY	STATUS	RESTARTS	AGE
gateway-service-85bd46cc4c-7zkh5	1/1	Running	0	43s
gateway-service-85bd46cc4c-fh7nf	1/1	Running	0	36s

Scaling — scale to 4, then back to 2:

```
$ kubectl scale deployment/gateway-service --replicas=4
deployment.apps/gateway-service scaled
```

```
$ kubectl get pods -l app=gateway-service
```

NAME	READY	STATUS	RESTARTS	AGE
gateway-service-85bd46cc4c-7zkh5	1/1	Running	0	50s
gateway-service-85bd46cc4c-fh7nf	1/1	Running	0	43s
gateway-service-85bd46cc4c-fp9lv	1/1	Running	0	7s
gateway-service-85bd46cc4c-phsm9	1/1	Running	0	7s


```
$ kubectl scale deployment/gateway-service --replicas=2 # extras Terminating
```

Load balancing — 6 requests to the Service answered by different Pods (the `instance` field is the serving Pod's hostname):

```
{
  "status": "ok",
  "service": "gateway",
  "instance": "gateway-service-85bd46cc4c-fh7nf"
}
{"status": "ok", "service": "gateway", "instance": "gateway-service-85bd46cc4c-fh7nf"}
{"status": "ok", "service": "gateway", "instance": "gateway-service-85bd46cc4c-7zkh5"}
{"status": "ok", "service": "gateway", "instance": "gateway-service-85bd46cc4c-fh7nf"}
{"status": "ok", "service": "gateway", "instance": "gateway-service-85bd46cc4c-7zkh5"}
{"status": "ok", "service": "gateway", "instance": "gateway-service-85bd46cc4c-fh7nf"}
```

4. Technical Guide — Part B: Active Redundancy (Hot Spare)

4.1 Pattern description

In **Active Redundancy (Hot Spare)** every node in the protection group — active *and* spare — receives and processes **identical inputs in parallel**, so the spare's state is fully synchronized with the active at all times. On failure the spare takes over with **no state loss**. It realizes the **Redundant Spare** tactic (Recover from Faults → Preparation and Repair).

In GameSeeker we apply it to the **ranking-service** (the live deal leaderboard). Previously a single ranking node consumed `ranking_prices_queue` — a single point of failure. Now:

- price events are published to a **fanout exchange**, so the active and spare each get *every* message on their own queue (parallel identical inputs);
- both write to **shared Redis**, so their leaderboards are identical and the spare is genuinely hot;
- a **coordinator** detects active failure and promotes the spare in ms.

4.2 Quality scenario (six-part)

Element	Value
Source	An internal fault — process crash / container failure of a ranking node.
Stimulus	The active <code>ranking-service</code> instance stops responding (crashes).
Artifact	The ranking subsystem (deal leaderboard read path).
Environment	Normal operation, under a live stream of price events from the scrapper.
Response	The coordinator detects the active is down via health checks and reroutes leaderboard reads to the hot spare, which already holds an identical leaderboard.
Response measure	Failover completes in well under one second (measured ≈ 0.5 s, dominated by fault detection; the state takeover itself is sub-millisecond) with zero leaderboard entries lost and no client-visible errors on subsequent reads.

4.3 Implementation steps

1. **Fanout topology.** `ranking-service` declares a fanout exchange and a per-instance queue bound to it; the `@RabbitListener` listens on that per-instance queue (`config/RabbitMQConfig.java` , `messaging/GamePriceConsumer.java` , `application.properties`).
2. **Publisher.** `scraper-service` publishes ranking events to the fanout exchange instead of a single queue (`scraper-service/src/repositories/brokers/rabbitmq.py`).
3. **Two nodes.** Run `ranking-active` and `ranking-spare` from the same image, differing only by `RANKING_INSTANCE_ID` / `RANKING_INSTANCE_QUEUE` , sharing one Redis.
4. **Coordinator.** A FastAPI service (`reliability/ranking-coordinator/`) health-checks both nodes every 250 ms, routes `/api/v1/ranking/*` reads to the primary, promotes the spare after 2 consecutive failures, and fails back when the active recovers. The gateway's `RANKING_SERVICE_URL` now points at the coordinator — the hot spare is transparent to the rest of the system.
5. **Run & demonstrate:**

```
docker compose -f reliability/docker-compose.reliability.yml up --build -d
bash reliability/failover-demo.sh
```

4.4 Configuration / code snippets

Fanout topology so both nodes get every message (state-sync mechanism):

```
// ranking-service/.../config/RabbitMQConfig.java (excerpt)
@Bean FanoutExchange rankingExchange() { return new FanoutExchange(exchangeName, true,
false); }
@Bean Queue rankingInstanceQueue() { return new Queue(instanceQueueName, false, false,
true); }
@Bean Binding rankingBinding() { return
BindingBuilder.bind(rankingInstanceQueue()).to(rankingExchange()); }
```

Coordinator promotion logic (health check → failover):

```
# reliability/ranking-coordinator/coordinator.py (excerpt)
primary_down = state["consecutive_failures"][primary] >= FAIL_THRESHOLD
if primary_down and state["healthy"][spare]:
    _promote(spare, "FAILOVER") # spare already hot -> zero state loss
elif FAILBACK and primary == "spare" and state["healthy"]["active"]:
    _promote("active", "FAILBACK")
```

4.5 Evidence — failover

Captured from reliability/failover-demo.sh .

1) Steady state — both nodes hot, leaderboards identical (15 = 15):

```
{ "primary": "active",
  "healthy": { "active": true, "spare": true },
  "leaderboard_size": { "active": 15, "spare": 15 },
  "failover_count": 0, "last_failover_ms": null }
```

Reads are served by the active node:

```
$ curl -D - .../api/v1/ranking/top?limit=5
x-served-by: active
x-failover-count: 0
```

2) Fault injection — stop the active node:

```
$ docker stop ranking-active
```

3) Failover — coordinator promotes the spare (~0.5 s, zero loss):

```
{ "primary": "spare",
  "healthy": { "active": false, "spare": true },
  "leaderboard_size": { "active": null, "spare": 15 },
  "failover_count": 1,
  "last_failover_ms": 518.6,
  "last_event": "FAILOVER: primary active -> spare (detected in ~518.6 ms)" }
```

Reads continue uninterrupted, now served by the spare — **same 15 entries, no state lost** (the spare had been processing every price event in parallel):

```
$ curl -D - - .../api/v1/ranking/top?limit=5
x-served-by: spare
```

4) Recovery — restart the active node, coordinator fails back:

```
{ "primary": "active",
  "healthy": { "active": true, "spare": true },
  "leaderboard_size": { "active": 15, "spare": 15 },
  "failover_count": 2,
  "last_event": "FAILBACK: primary spare -> active" }
```

4.6 Recommendations (for teams adopting this pattern)

1. **Use a fanout (or topic) exchange, not a shared queue.** With a single shared queue the two instances become *competing consumers* (each message goes to only one) — that is load-sharing, not redundancy. A fanout gives each node its own copy, which is what keeps the spare hot.
2. **Externalize or fully replicate state.** We put leaderboard state in shared Redis so promotion is instantaneous. If you keep per-node state instead, make absolutely sure every input reaches every node, and verify convergence (the coordinator's `/status` reports each node's leaderboard size for exactly this).
3. **Separate detection latency from failover latency, and tune it.** The takeover itself is sub-millisecond (a variable flip) because the spare is already synchronized; the user-visible delay is dominated by *detection* (`HEALTH_INTERVAL × FAIL_THRESHOLD`). Keep the threshold ≥ 2 to avoid flapping on transient blips, and size the interval to your latency budget.
4. **Make the spare transparent.** Putting the coordinator behind the same API/URL the gateway already used meant zero changes elsewhere — adopt the redundancy behind an existing seam rather than spreading failover logic across callers.

5. Pull Request

Implementation branch: **reliability** → merged into `main` and delivered as a Pull Request toward the **prototype_3** branch of the course repository.

6. Reproducing everything locally

```
# Part A – Cluster Pattern (Kubernetes)
bash k8s/demo.sh

# Part B – Active Redundancy / Hot Spare
docker compose -f reliability/docker-compose.reliability.yml up --build -d
bash reliability/failover-demo.sh
docker compose -f reliability/docker-compose.reliability.yml down
```